

Chapter 5

Feedforward Networks as Spin Systems:

A Replica Calculation in Weight Space

Physics of Learning • IIT Madras

In Chapter 4 we applied the replica method to a single binary spin carrying a random field. Here we ask whether the same spin-glass structure arises in a feedforward neural network. The answer is yes, but the “spins” are now the *weights*, not the neurons, and the quenched disorder is the *training data*, not fixed couplings. We make this precise on the simplest possible feedforward model—a two-layer network with one neuron per layer—and carry out the replica calculation in full, showing every algebraic step. The calculation is presented explicitly as a pedagogical device; the full- N literature is cited at the close.

The Minimal Two-Layer MLP

WE WORK WITH THE SMALLEST possible feedforward architecture: one input neuron, one hidden neuron, one output neuron, connected by two scalar weights.

Architecture and forward pass

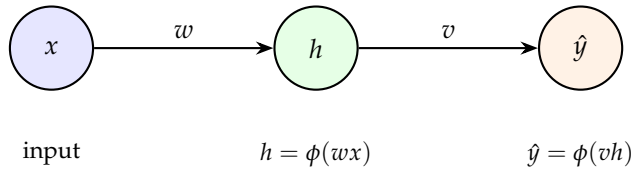


Figure 1: The minimal two-layer MLP. There are exactly two scalar weights: w (input-to-hidden) and v (hidden-to-output). The configuration of this system is the pair $(w, v) \in \mathbb{R}^2$.

The forward pass is

$$h^\mu = \phi(wx^\mu), \quad (1)$$

$$\hat{y}^\mu = \phi(v h^\mu) = \phi(v \phi(wx^\mu)). \quad (2)$$

We work in the **linear regime**: weights are small enough that $\phi(z) \approx z$. Then

$$h^\mu \approx wx^\mu, \quad \hat{y}^\mu \approx v \cdot wx^\mu = u x^\mu, \quad (3)$$

where we define the **composite weight** $u \equiv vw$. The two-parameter model collapses to a single effective parameter, making every integral explicit.

Training data

We have P i.i.d. samples

$$x^\mu \sim \mathcal{N}(0,1), \quad y^\mu = \text{sign}(x^\mu) \in \{-1, +1\}, \quad \mu = 1, \dots, P. \quad (4)$$

The training data are drawn once and held fixed throughout; they are **quenched disorder**, playing the role of the random couplings J_{ij} in a spin glass.

Loss function as Hamiltonian

The mean-squared-error loss is

$$\mathcal{L}(u; \mathcal{D}) = \frac{1}{P} \sum_{\mu=1}^P (y^\mu - u x^\mu)^2. \quad (5)$$

This is the **Hamiltonian** of the system: a function of the “spin” u (the weight) for fixed disorder $\mathcal{D}$ (the data).

Expanding the square. Write out $(y^\mu - u x^\mu)^2 = (y^\mu)^2 - 2u y^\mu x^\mu + u^2 (x^\mu)^2$ and sum over μ :

$$\mathcal{L}(u) = \frac{1}{P} \sum_{\mu} (y^\mu)^2 - 2u \cdot \frac{1}{P} \sum_{\mu} y^\mu x^\mu + u^2 \cdot \frac{1}{P} \sum_{\mu} (x^\mu)^2. \quad (6)$$

Since $y^\mu = \pm 1$ we have $(y^\mu)^2 = 1$, so the first sum equals 1. Define

$$\hat{C} \equiv \frac{1}{P} \sum_{\mu=1}^P y^\mu x^\mu, \quad \hat{Q} \equiv \frac{1}{P} \sum_{\mu=1}^P (x^\mu)^2. \quad (7)$$

By the law of large numbers $\hat{Q} \rightarrow \mathbb{E}[(x^\mu)^2] = 1$ as $P \rightarrow \infty$.¹ The loss becomes

$$\boxed{\mathcal{L}(u) = 1 - 2u\hat{C} + u^2.} \quad (8)$$

¹ We use $\hat{Q} \approx 1$ henceforth; corrections are $O(1/\sqrt{P})$.

The only random quantity is $\hat{C}$, the **signal correlation** between the inputs and the labels. It is fixed once the dataset is drawn—a random external field acting on the composite weight u .

Partition function in weight space

Assigning a Boltzmann weight at inverse temperature β to each weight configuration, the partition function is

$$Z(\mathcal{D}) = \int_{-\infty}^{\infty} du e^{-\beta \mathcal{L}(u)}. \quad (9)$$

Statistical mechanics $\leftrightarrow$ **Bayesian inference**. Equation (9) is the normalisation of a Bayesian posterior over weights with a flat (improper) prior, where β controls the sharpness of the loss. At $\beta \rightarrow \infty$ the distribution collapses onto the loss-minimising weight—the trained network. At finite β , “thermal” fluctuations represent uncertainty in the weight given finite data. The free energy $F = -\frac{1}{\beta} \ln Z$ encodes this uncertainty.

Substituting (8) into (9):

$$\begin{aligned} Z(\mathcal{D}) &= \int du \exp(-\beta(1 - 2u\hat{C} + u^2)) \\ &= e^{-\beta} \int du \exp(-\beta u^2 + 2\beta\hat{C}u). \end{aligned} \quad (10)$$

This is a Gaussian integral. Complete the square in the exponent:

$$-\beta u^2 + 2\beta\hat{C}u = -\beta(u - \hat{C})^2 + \beta\hat{C}^2. \quad (11)$$

Using $\int_{-\infty}^{\infty} e^{-\beta(u-\hat{C})^2} du = \sqrt{\pi/\beta}$, the partition function evaluates exactly to

$$Z(\mathcal{D}) = \sqrt{\frac{\pi}{\beta}} \exp(\beta\hat{C}^2 - \beta). \quad (12)$$

This is the exact, closed-form partition function for a fixed dataset. The free energy is therefore $F(\mathcal{D}) = -\frac{1}{\beta} \ln Z = 1 - \hat{C}^2 - \frac{1}{2\beta} \ln \frac{\pi}{\beta}$, which depends on the random quantity $\hat{C}$ and hence varies from dataset to dataset.

The Quenched Free Energy via the Replica Trick

THE PHYSICALLY MEANINGFUL QUANTITY is the free energy *averaged over all datasets*—the **quenched free energy**:

$$f \equiv -\frac{1}{\beta} \mathbb{E}_{\mathcal{D}}[\ln Z(\mathcal{D})]. \quad (13)$$

The difficulty is identical to Chapter 4: we must average a logarithm over the disorder. The **replica trick** converts this into a limit of integer moments,

$$\mathbb{E}[\ln Z] = \lim_{n \rightarrow 0} \frac{\mathbb{E}[Z^n] - 1}{n} = \lim_{n \rightarrow 0} \frac{1}{n} \ln \mathbb{E}[Z^n], \quad (14)$$

and the strategy is: (i) compute $\mathbb{E}[Z^n]$ for integer n , (ii) identify the closed-form expression, (iii) take $n \rightarrow 0$ by analytic continuation.

Step 1 — Introduce n replicas

Z^n is the product of n independent copies of the partition function, each with its own weight u^a ($a = 1, \dots, n$), but **all sharing the same dataset $\mathcal{D}$** :

$$Z^n(\mathcal{D}) = \prod_{a=1}^n \left[\int du^a e^{-\beta \mathcal{L}(u^a)} \right] = \int \prod_{a=1}^n du^a \exp \left(-\beta \sum_{a=1}^n \mathcal{L}(u^a) \right). \quad (15)$$

Substituting (8) into the sum over replicas:

$$\begin{aligned} \sum_{a=1}^n \mathcal{L}(u^a) &= \sum_a \left(1 - 2u^a \hat{C} + (u^a)^2 \right) \\ &= n - 2\hat{C} \sum_a u^a + \sum_a (u^a)^2. \end{aligned} \quad (16)$$

Therefore

$$Z^n(\mathcal{D}) = e^{-n\beta} \int \prod_a du^a \exp \left(-\beta \sum_a (u^a)^2 + 2\beta \hat{C} \sum_a u^a \right). \quad (17)$$

The factor $\exp(2\beta \hat{C} \sum_a u^a)$ ties all replicas together through the shared dataset. This **inter-replica coupling** is the hallmark of the spin-glass problem in weight space.

Step 2 — Average over the disorder

We now compute $\mathbb{E}_{\mathcal{D}}[Z^n]$ by averaging over $\hat{C}$.

Distribution of $\hat{C}$. The signal correlation $\hat{C} = \frac{1}{P} \sum_{\mu} y^{\mu} x^{\mu}$ is a sum of P i.i.d. random variables. Their mean is

$$\mathbb{E}[y^{\mu} x^{\mu}] = \mathbb{E}[\text{sign}(x) \cdot x] = 0, \quad (18)$$

which follows from the symmetry $x \mapsto -x$ of $\mathcal{N}(0, 1)$, since $\text{sign}(-x)(-x) = \text{sign}(x)x$.² Their second moment is

$$\mathbb{E}[(y^{\mu} x^{\mu})^2] = \mathbb{E}[(x^{\mu})^2] = 1, \quad (19)$$

since $(y^{\mu})^2 = 1$ always. By the central limit theorem

$$\hat{C} \sim \mathcal{N}\left(0, \frac{1}{P}\right). \quad (20)$$

Averaging the coupling term. We need $\mathbb{E}_{\hat{C}}[\exp(2\beta \hat{C} \sum_a u^a)]$. For any $X \sim \mathcal{N}(0, \sigma^2)$ the moment generating function is

$$\mathbb{E}[e^{\lambda X}] = e^{\lambda^2 \sigma^2 / 2}. \quad (21)$$

² More explicitly: $\mathbb{E}[\text{sign}(x)x] = \int_0^{\infty} x p(x) dx + \int_{-\infty}^0 (-x)(-p(x)) dx = 2 \int_0^{\infty} x p(x) dx = 0$ by symmetry of $p(x)$.

Setting $X = \hat{C}$, $\sigma^2 = 1/P$, and $\lambda = 2\beta \sum_a u^a$:

$$\begin{aligned} \mathbb{E}_{\hat{C}} \left[\exp \left(2\beta \hat{C} \sum_a u^a \right) \right] &= \exp \left(\frac{(2\beta \sum_a u^a)^2}{2P} \right) \\ &= \exp \left(\frac{2\beta^2}{P} \left(\sum_a u^a \right)^2 \right). \end{aligned} \quad (22)$$

Expanding the square. Using $(\sum_a u^a)^2 = \sum_a (u^a)^2 + 2 \sum_{a < b} u^a u^b$:

$$\frac{2\beta^2}{P} \left(\sum_a u^a \right)^2 = \frac{2\beta^2}{P} \sum_a (u^a)^2 + \frac{4\beta^2}{P} \sum_{a < b} u^a u^b. \quad (23)$$

Substituting into (17) and incorporating both $-\beta \sum_a (u^a)^2$ from (17) and the two new terms:

$$\mathbb{E}[Z^n] = e^{-n\beta} \int \prod_a du^a \exp \left(- \underbrace{\left(\beta - \frac{2\beta^2}{P} \right) \sum_a (u^a)^2}_{\equiv \beta_{\text{eff}}} + \frac{4\beta^2}{P} \sum_{a < b} u^a u^b \right). \quad (24)$$

The off-diagonal term $\frac{4\beta^2}{P} \sum_{a < b} u^a u^b$ couples different replicas. It *prevents* the integral over $\prod_a du^a$ from factorising. We must remove it.

Step 3 — Identify the order parameter

Before decoupling, we note what the coupling coefficient measures. Define the **replica overlap**

$$q^{ab} \equiv \mathbb{E}_{\mathcal{D}} [\langle u^a \rangle \langle u^b \rangle], \quad a \neq b, \quad (25)$$

the correlation between the mean weights of two independently trained networks on the same dataset. This is the weight-space analogue of the **Edwards–Anderson order parameter**:

$q = 1$ means the dataset completely pins the weight; $q = 0$ means different training runs find uncorrelated solutions. The replica calculation will tell us the value of q as a function of P and β .

Under the **replica-symmetric (RS) ansatz** we assume $q^{ab} = q$ for all $a \neq b$ —all pairs of replicas overlap equally. This is the assumption that the weight-space Gibbs measure has a single basin, which is appropriate when the loss landscape is convex (as it is here in the linear regime).

Step 4 — Hubbard-Stratonovich decoupling

To factorise the replicas we use the **Hubbard-Stratonovich (HS) identity**: for any $A > 0$,

$$e^{A(\sum_a u^a)^2} = \sqrt{\frac{1}{4\pi A}} \int_{-\infty}^{\infty} dz \exp\left(-\frac{z^2}{4A} + z \sum_a u^a\right). \quad (26)$$

Proof of (26). On the right-hand side, complete the square in z :

$$-\frac{z^2}{4A} + z \sum_a u^a = -\frac{1}{4A} \left(z - 2A \sum_a u^a\right)^2 + A \left(\sum_a u^a\right)^2. \quad (27)$$

The Gaussian integral over z then gives $\int dz \exp\left(-\frac{1}{4A}(z - 2A \sum_a u^a)^2\right) = \sqrt{4\pi A}$, and

$$\text{RHS} = \sqrt{\frac{1}{4\pi A}} \cdot \sqrt{4\pi A} \cdot e^{A(\sum_a u^a)^2} = e^{A(\sum_a u^a)^2} = \text{LHS}. \quad (28)$$

We apply (26) to the coupling term in (24). We need to rewrite $\exp\left(\frac{2\beta^2}{P}(\sum_a u^a)^2\right)$ using the HS identity with $A = 2\beta^2/P$, which gives $\frac{1}{4A} = \frac{P}{8\beta^2}$:

$$\exp\left(\frac{2\beta^2}{P} \left(\sum_a u^a\right)^2\right) = \sqrt{\frac{P}{8\pi\beta^2}} \int dz \exp\left(-\frac{Pz^2}{8\beta^2} + z \sum_a u^a\right). \quad (29)$$

But note from (24) that the coupling term is $\exp(\frac{4\beta^2}{P} \sum_{a<b} u^a u^b)$, not $\exp(\frac{2\beta^2}{P}(\sum_a u^a)^2)$ directly. These differ by a diagonal piece. We reconcile them by writing

$$\frac{4\beta^2}{P} \sum_{a<b} u^a u^b = \frac{2\beta^2}{P} \left(\sum_a u^a\right)^2 - \frac{2\beta^2}{P} \sum_a (u^a)^2, \quad (30)$$

so that (24) becomes

$$\mathbb{E}[Z^n] = e^{-n\beta} \int \prod_a du^a \exp\left(-\beta \sum_a (u^a)^2\right) \cdot \exp\left(\frac{2\beta^2}{P} \left(\sum_a u^a\right)^2\right). \quad (31)$$

(The $+\frac{2\beta^2}{P} \sum_a (u^a)^2$ from expanding the square and the $-\frac{2\beta^2}{P} \sum_a (u^a)^2$ from (30) cancel, leaving a clean $-\beta \sum_a (u^a)^2$.) Now apply the HS identity (29):

$$\mathbb{E}[Z^n] = e^{-n\beta} \sqrt{\frac{P}{8\pi\beta^2}} \int dz e^{-Pz^2/(8\beta^2)} \int \prod_a du^a \exp\left(-\beta \sum_a (u^a)^2 + z \sum_a u^a\right). \quad (32)$$

The exponent now separates across replicas: $-\beta \sum_a (u^a)^2 + z \sum_a u^a = \sum_a (-\beta (u^a)^2 + z u^a)$. The integral over $\prod_a du^a$ therefore factorises into n identical single-replica integrals:

$$\int \prod_a du^a e^{-\beta \sum_a (u^a)^2 + z \sum_a u^a} = \left[\int du e^{-\beta u^2 + zu} \right]^n. \quad (33)$$

Step 5 — Evaluate the single-replica integral

Each single-replica integral is Gaussian. Complete the square in u :

$$-\beta u^2 + zu = -\beta \left(u - \frac{z}{2\beta} \right)^2 + \frac{z^2}{4\beta}. \quad (34)$$

Then

$$\begin{aligned} \int du e^{-\beta u^2 + zu} &= e^{z^2/(4\beta)} \int du e^{-\beta (u - z/(2\beta))^2} \\ &= \sqrt{\frac{\pi}{\beta}} e^{z^2/(4\beta)}. \end{aligned} \quad (35)$$

Substituting (35) and (33) into (32):

$$\mathbb{E}[Z^n] = e^{-n\beta} \sqrt{\frac{P}{8\pi\beta^2}} \left(\frac{\pi}{\beta} \right)^{n/2} \int dz \exp\left(-\frac{Pz^2}{8\beta^2} + \frac{nz^2}{4\beta}\right). \quad (36)$$

Step 6 — Evaluate the z -integral

Collect the exponent in z :

$$-\frac{Pz^2}{8\beta^2} + \frac{nz^2}{4\beta} = -\frac{z^2}{2} \left(\frac{P}{4\beta^2} - \frac{n}{2\beta} \right). \quad (37)$$

This is a Gaussian in z with variance $\sigma_z^2 = \left(\frac{P}{4\beta^2} - \frac{n}{2\beta} \right)^{-1}$. The integral evaluates to

$$\int dz \exp\left(-\frac{z^2}{2} \left(\frac{P}{4\beta^2} - \frac{n}{2\beta} \right)\right) = \sqrt{\frac{2\pi}{\frac{P}{4\beta^2} - \frac{n}{2\beta}}}. \quad (38)$$

Inserting into (36):

$$\mathbb{E}[Z^n] = e^{-n\beta} \sqrt{\frac{P}{8\pi\beta^2}} \left(\frac{\pi}{\beta} \right)^{n/2} \sqrt{\frac{2\pi}{\frac{P}{4\beta^2} - \frac{n}{2\beta}}}. \quad (39)$$

Step 7 — Take $\ln$ and expand in n

Taking $\ln$ of (39) and separating terms by their dependence on n :

$$\ln \mathbb{E}[Z^n] = -n\beta + \frac{1}{2} \ln \frac{P}{8\pi\beta^2} + \frac{n}{2} \ln \frac{\pi}{\beta} + \frac{1}{2} \ln \frac{2\pi}{\frac{P}{4\beta^2} - \frac{n}{2\beta}}. \quad (40)$$

The last term depends on n . Expand it around $n = 0$:

$$\begin{aligned} \frac{1}{2} \ln \frac{2\pi}{\frac{P}{4\beta^2} - \frac{n}{2\beta}} &= \frac{1}{2} \ln \frac{8\pi\beta^2}{P} + \frac{1}{2} \ln \frac{1}{1 - \frac{2n\beta}{P}} \\ &= \frac{1}{2} \ln \frac{8\pi\beta^2}{P} + \frac{1}{2} \cdot \frac{2n\beta}{P} + O(n^2) \\ &= \frac{1}{2} \ln \frac{8\pi\beta^2}{P} + \frac{n\beta}{P} + O(n^2), \end{aligned} \quad (41)$$

where we used $\ln(1 - \epsilon) \approx -\epsilon$ for small $\epsilon = 2n\beta/P$. Substituting (41) into (40):

$$\begin{aligned} \ln \mathbb{E}[Z^n] &= -n\beta + \frac{1}{2} \ln \frac{P}{8\pi\beta^2} + \frac{n}{2} \ln \frac{\pi}{\beta} + \frac{1}{2} \ln \frac{8\pi\beta^2}{P} + \frac{n\beta}{P} + O(n^2) \\ &= n \left(-\beta + \frac{1}{2} \ln \frac{\pi}{\beta} + \frac{\beta}{P} \right) + O(n^2), \end{aligned} \quad (42)$$

where the two constant (in n) terms $\frac{1}{2} \ln \frac{P}{8\pi\beta^2}$ and $\frac{1}{2} \ln \frac{8\pi\beta^2}{P}$ cancel exactly.

Step 8 — Take $n \rightarrow 0$

By the replica trick (14):

$$\begin{aligned} \mathbb{E}[\ln Z] &= \lim_{n \rightarrow 0} \frac{1}{n} \ln \mathbb{E}[Z^n] = \lim_{n \rightarrow 0} \frac{1}{n} \left[n \left(-\beta + \frac{1}{2} \ln \frac{\pi}{\beta} + \frac{\beta}{P} \right) + O(n^2) \right] \\ &= -\beta + \frac{1}{2} \ln \frac{\pi}{\beta} + \frac{\beta}{P}. \end{aligned} \quad (43)$$

The **quenched free energy per sample** is therefore

$$\boxed{f = -\frac{1}{\beta} \mathbb{E}[\ln Z] = 1 - \frac{1}{P} - \frac{1}{2\beta} \ln \frac{\pi}{\beta}.} \quad (44)$$

The Order Parameter

THE REPLICAS OVERLAP q can be extracted from the same calculation without further work. In Step 5, the mean of u in the single-replica integral conditioned on the HS field z is

$$\langle u \rangle_z = \frac{z}{2\beta}, \quad (45)$$

which follows from differentiating $\ln \int du e^{-\beta u^2 + zu}$ with respect to z . Two replicas a and b , independently integrated but conditioned on the same z (i.e. the same dataset), have means $\langle u^a \rangle_z = \langle u^b \rangle_z = z/(2\beta)$. Their overlap is therefore

$$q^{ab} = \mathbb{E}_z \left[\langle u^a \rangle_z \langle u^b \rangle_z \right] = \mathbb{E}_z \left[\frac{z^2}{4\beta^2} \right]. \quad (46)$$

At $n = 0$, the HS field z is distributed as $z \sim \mathcal{N}(0, \sigma_z^2)$ with $\sigma_z^2 = (P/(4\beta^2))^{-1} = 4\beta^2/P$. Therefore

$$\mathbb{E}_z \left[\frac{z^2}{4\beta^2} \right] = \frac{1}{4\beta^2} \cdot \frac{4\beta^2}{P} = \frac{1}{P}. \quad (47)$$

Edwards–Anderson order parameter in weight space:

$$q = \frac{1}{P}.$$

Two networks trained independently on the *same* P -sample dataset have weight overlap $q = 1/P$. As $P \rightarrow \infty$, $q \rightarrow 0$: with abundant data, independently trained networks find uncorrelated weight solutions (the loss landscape is flat and the solution space is broad). As $P \rightarrow 0$, q diverges, signalling that even a tiny dataset sharply pins the weight.

Reading the Physics

THE FREE ENERGY (44) has three pieces, each with a clear physical interpretation.

Baseline loss: the constant 1. This is the MSE incurred by a zero-weight predictor $\hat{y} = 0$, since $\mathbb{E}[(y^\mu)^2] = 1$. It is the starting point before the data have any influence.

Data correction: $-1/P$. This is the reduction in free energy due to having P training samples. It vanishes as $P \rightarrow \infty$, consistent with the weight being well-determined by abundant data.

Entropic term: $-\frac{1}{2\beta} \ln \frac{\pi}{\beta}$. This is the contribution from the volume of the weight distribution in the absence of data. At high temperature (small β) entropy dominates; the weights are broadly distributed and the free energy is large and positive. At low temperature (large β) the network concentrates on the loss minimum and this term becomes negligible.

Cross-check. We can verify (44) against the exact result (12). From (12), $\ln Z = \beta \hat{C}^2 - \beta + \frac{1}{2} \ln \frac{\pi}{\beta}$. Taking the disorder average and using $\mathbb{E}[\hat{C}^2] = \text{Var}(\hat{C}) = 1/P$ from (20):

$$\mathbb{E}[\ln Z] = \beta \cdot \frac{1}{P} - \beta + \frac{1}{2} \ln \frac{\pi}{\beta} \quad (48)$$

which matches (43) exactly. ✓

This cross-check is a luxury available only because the toy model is simple enough to yield a closed-form $Z(\mathcal{D})$. In the full N -dimensional

problem the replica method is the *only* available route; there is no exact Z to compare against.

The Spin-Glass Correspondence

THE ENTIRE CALCULATION maps onto the SK spin glass with the following dictionary.

SK Spin Glass	Two-layer MLP
Spin configuration $\{s_i\}$	Composite weight $u = vw$
Hamiltonian $H(\mathbf{s})$	Loss function $\mathcal{L}(u)$
Quenched couplings J_{ij}	Training data (x^μ, y^μ)
Partition function $\sum e^{-\beta H}$	$\int du e^{-\beta \mathcal{L}}$
Disorder average $\mathbb{E}_J[\cdot]$	Dataset average $\mathbb{E}_{\mathcal{D}}[\cdot]$
Free energy $F = -\frac{1}{\beta} \ln Z$	Log marginal likelihood
Temperature $T = 1/\beta$	Inverse loss sharpness
Edwards–Anderson parameter q	Inter-replica weight overlap
Replica-symmetric phase	Single-basin loss landscape
Replica-symmetry breaking	Glassy, multi-basin landscape

Table 1: Correspondence between SK spin-glass language and the statistical mechanics of the two-layer MLP.

The key structural difference from the Hopfield network is this: there, the spins *are* the neurons and the disorder *is* the couplings; the spin-glass structure lives in neuron configuration space. Here, the spins are the *weights* and the disorder is the *data*; the spin-glass structure lives in weight space. The replica machinery is identical; only the interpretation of the variables changes.

A Note on the Literature

THE SCALAR TOY MODEL presented here does not appear as a standalone result in the literature. It is a deliberate simplification in which the thermodynamic limit $N \rightarrow \infty$ (essential for sharpness of the physics) is replaced by $N = 1$ to make every step traceable by hand.

The calculations in the literature that this toy model illustrates are:

- **Gardner (1988).** The foundational replica calculation for the *single-layer* perceptron with N -dimensional weight vectors. By computing the volume of weight space consistent with classifying $P = \alpha N$ random patterns correctly, Gardner derives the storage capacity $\alpha_c = 2$. The steps are the N -dimensional version of Steps 1–8 above, with the HS field becoming a scalar that is then solved by a saddle-point equation.

- **Engel, Köhler et al. (1992).** Extension of Gardner’s calculation to a two-layer feedforward network (the committee machine) with N inputs and K hidden units. The replica-symmetric result is $\alpha_c \propto \sqrt{K}$.
- **Engel & van den Broeck (2001).** A comprehensive textbook treatment covering perceptron learning, replica calculations, and the connection to generalisation—the natural next reference after this chapter.
- **Bahri, Kadmon, Pennington et al. (2020).** Modern treatment of high-dimensional learning at large and infinite width, covering phase transitions in generalisation, and the Neural Tangent Kernel regime.

Chapter 6 takes the next step: the SK model in the thermodynamic limit, where the same replica machinery yields genuine phase transitions, the Parisi order parameter function, and the derivation of $\alpha_c \approx 0.138$ for the Hopfield associative memory.

References. E. Gardner, *J. Phys. A: Math. Gen.* **21**, 257 (1988). A. Engel, H.M. Köhler, F. Tschepeke, H. Vollmayr, A. Zippelius, *Phys. Rev. A* **45**, 7590 (1992). A. Engel & C. van den Broeck, *Statistical Mechanics of Learning*, Cambridge University Press (2001). Y. Bahri, J. Kadmon, J. Pennington, S.S. Schoenholz, J. Sohl-Dickstein, S. Ganguli, *Annu. Rev. Condens. Matter Phys.* **11**, 501 (2020).